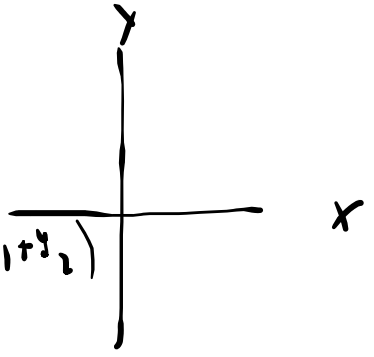
$-1, 0, 1, 2, 3$ $+$ Group

$+ \times$ Field

Complex – number

$z = (x, y)$



Add: $z_1 + z_2 = (x_1, y_1) + (x_2, y_2) = (x_1 + x_2, y_1 + y_2)$

Mul. $z_1 z_2 = (x_1 x_2 - y_1 y_2, x_1 y_2 + x_2 y_1)$

$i = (0,1)$

$i^2 = (0,1)(0,1) = (-1,0) = -1$

$z = (x,y) = (x,0) + (0,y) = \boxed{x + iy}$ (real, imaginary)

Ex $(2+3i)(1-2i) = (2+6) - i = 8 - i$

Complex conjugate $z^* = x - iy$

$zz^* = (x+iy)(x-iy) = x^2 + y^2$

$|z| = \sqrt{zz^*}$

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Complex plane $\equiv \mathbb{C}$

Ex $\dfrac{z_1}{z_2}\dfrac{z_2^*}{z_2^*} = \dfrac{z_1 z_2^*}{x_2^2+y_2^2} = \dfrac{x_1x_2+y_1y_2+(x_2y_1-x_1y_2)i}{x_2^2+y_2^2}$

Functions $z = x+iy$

$f(z) = u(x,y) + i\,v(x,y)$

Real function $\longrightarrow$ Complex

Power series $\longrightarrow$ Converge

Ex $e^z = 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \cdots$

Ex $e^{iz} = 1 + (iz) + \frac{(iz)^2}{2!} + \frac{(iz)^3}{3!} + \frac{(iz)^4}{4!} + \cdots$

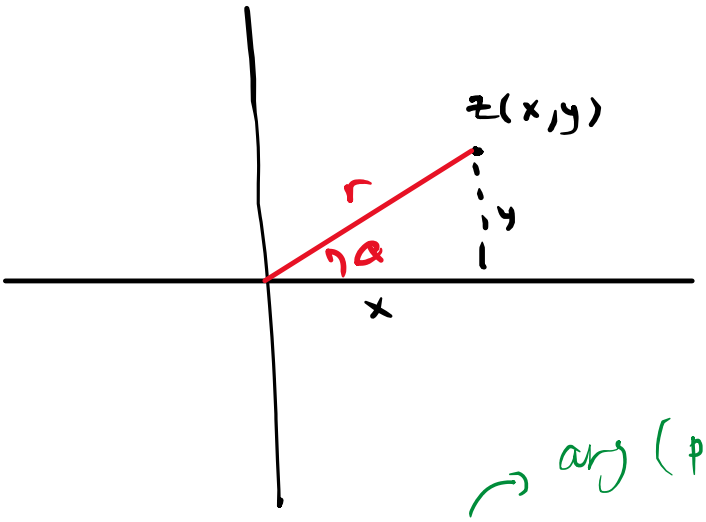
$= \left(1 - \frac{z^2}{2!} + \frac{z^4}{4!} - \cdots\right) + i\left(z - \frac{z^3}{3!} + \frac{z^5}{5!} - \cdots\right)$

$e^{iz} = \cos z + i \sin z$

Ex $f(z) = z^2 = (x+iy)^2 = (x^2 - y^2) + i(2xy)$

$u(x,y) = x^2 - y^2 \qquad v(x,y) = 2xy$

Polar rep. (Argand)



$r = \sqrt{zz^*}$

$x = r\cos\theta$

$y = r\sin\theta$

$z = r\cos\theta + ir\sin\theta$

$= r[\cos\theta + i\sin\theta] \Rightarrow z = re^{i\theta}$ → arg (phase)

↓

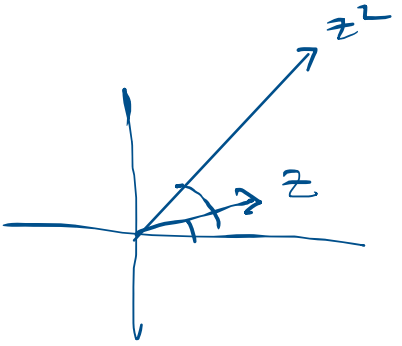
modulus $= |z|$

Multiply $z z' = r e^{i\theta} (r' e^{i\theta'})$

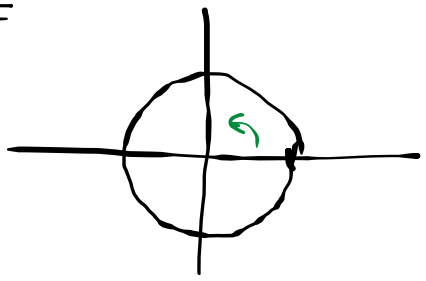
$= r r' e^{i(\theta + \theta')}$

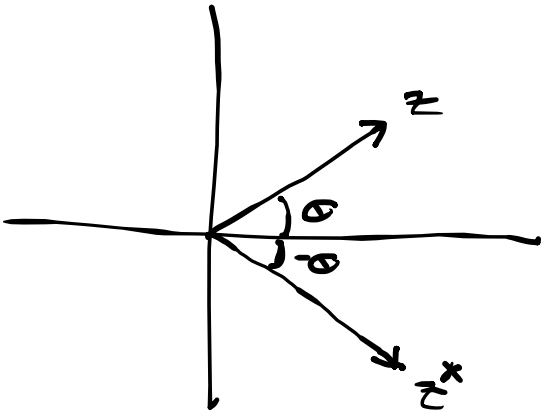
$\frac{z}{z'} = \frac{r}{r'} e^{i(\theta - \theta')}$

Ex $z^2 = r^2 e^{i 2\theta}$



1 $e^{i\alpha}$





$e^{i0} = 1$

$e^{i\frac{\pi}{2}} = +i$

$e^{i\pi} = -1$

$e^{i\frac{3\pi}{2}} = -i$

$\boxed{e^{i\omega t}}$

$e^{i(\omega t+\delta)}$

Ex $e^{i\theta} = \cos\theta + i\sin\theta$

de Moivre's Formula

$e^{in\theta} = \cos n\theta + i\sin n\theta = (\cos\theta + i\sin\theta)^n$

$\cos 2\theta + i\sin 2\theta = (\cos\theta + i\sin\theta)^2$

$= (\cos^2\theta - \sin^2\theta) + i(2\sin\theta\cos\theta)$

Circular & Hyper. Functions

$$\cos\alpha = \frac{e^{i\alpha} + e^{-i\alpha}}{2} \qquad \sin\alpha = \frac{e^{i\alpha} - e^{-i\alpha}}{2i}$$

$$\cosh\alpha = \frac{e^{\alpha} + e^{-\alpha}}{2} \qquad \sinh\alpha = \frac{e^{\alpha} - e^{-\alpha}}{2}$$

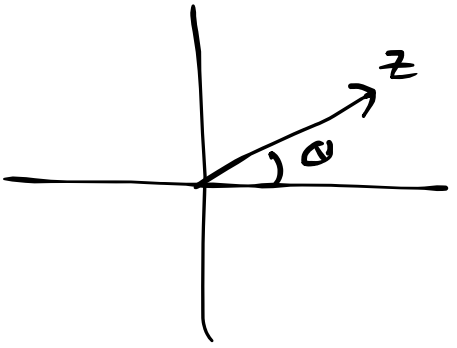
$$\cos z = \cosh(iz) \qquad i\sin z = \sinh(iz)$$

Power & root

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$z = r e^{i\alpha}$ $\qquad z^n = r^n e^{in\alpha}$

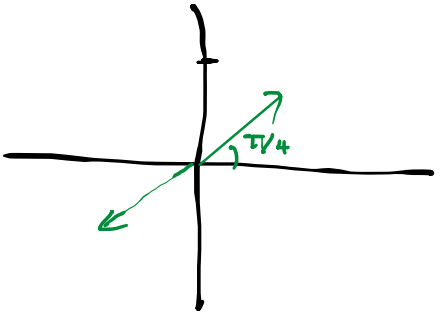
$z = r e^{i(\alpha + 2\pi m)}$



$z^{1/n} = r^{1/n} e^{i\left(\frac{\alpha}{n} + \frac{2\pi m}{n}\right)}$

Ex $\qquad z = i = 1\, e^{i\left(\frac{\pi}{2} + 2\pi m\right)}$

$z^{1/2} = 1\, e^{i\left(\frac{\pi}{4} + \frac{2\pi m}{2}\right)}$



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$$z = r e^{i(\theta + 2\pi n)}$$

$$\ln z = \underline{\ln r} + i(\theta + 2m\pi)$$

11.2 Derivative

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Derivative of $f(z) = u + iv$

$$\lim_{\delta z \to 0} \frac{f(z+\delta z) - f(z)}{\delta z}$$ independent of approach

$\delta z = \delta x + i\delta y$ $\quad \delta f = f(z+\delta z) - f(z) = \delta u + i\,\delta v$

Horizontal $\lim_{\delta z \to 0}$ $\quad \frac{\delta f}{\delta z} = \lim_{\delta x \to 0} \frac{\delta f}{\delta x} = \lim_{\delta x \to} \frac{\delta u}{\delta x} + i \lim_{\delta x \to} \frac{\delta v}{\delta x}$

$$\frac{\partial f}{\partial x} = \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x}$$

Vertical $\lim_{\delta z \to} \frac{\delta f}{\delta z} = \lim_{\delta y \to} \frac{\delta f}{i\delta y} = +\frac{\partial v}{\partial y} - i\frac{\partial u}{\partial y}$

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \& \quad \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}$$

Cauchy- Riemann Conditions (CRC)

$\frac{df}{dz} \Rightarrow$ CRC

CRC $\Rightarrow \frac{df}{dz}$??

$$\delta f = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy = \left(\frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}\right) dx + \left(\frac{\partial u}{\partial y} + i \frac{dv}{dy}\right) dy$$

$$= \left(\frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}\right) dx + \left(-\frac{\partial v}{\partial x} + i \frac{du}{dx}\right) dy$$

$$= \left(\frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}\right) dx + i\left(i \frac{\partial v}{\partial x} + \frac{du}{dx}\right) dy$$

$$= \left(\frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}\right)(dx + i\,dy) = \frac{\delta f}{\delta z} dz$$

$$\frac{\delta f}{\delta z} = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} \quad \checkmark$$

CRC & u & v continuous $\Rightarrow$ $f(z)$ analytic

regular

"holomorphic"

entire

$f'(z)$ doesn't exist at z_0 (singular point)

Ex $z = x + iy$

$\frac{\partial u}{\partial x} = 1$ ✓ $\quad \frac{\partial v}{\partial y} = 1$ ✓

$\frac{\partial u}{\partial y} = 0$ $\quad \frac{\partial v}{\partial x} = 0$ ✓

$z^2 = (x^2 - y^2) + i(2xy)$

$\frac{\partial u}{\partial x} = 2x$ $\quad \frac{\partial v}{\partial y} = 2x$ ✓

$\frac{\partial u}{\partial y} = -2y$ $\quad \frac{\partial v}{\partial x} = 2y$ ✓

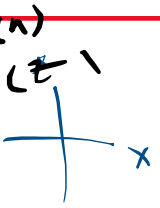
$P(z) = a z_0 + b z + c z^2 + \cdots$ analytic

Ex $\bar{z} = x - iy$ ✗

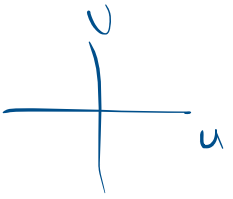
$$\text{CRC} \iff \frac{df}{dz^*} = 0$$

Properties of Analytic functions

* $f'(z)$ exists * All higher $f^{(n)}(z)$



$f(z)$ →



* Both parts u & v satisfy 2D Laplace equation:

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \qquad \frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0$$

"Harmonic functions"

$u(x,y) = \text{const} \Rightarrow du = 0$

$$\frac{\partial u}{\partial x}dx + \frac{\partial u}{\partial y}dy = 0$$

$$\left(\frac{dy}{dx}\right)_u = \frac{-\left(\frac{\partial u}{\partial x}\right)}{\left(\frac{\partial u}{\partial y}\right)}$$

$V(x,y) = \text{const} \Rightarrow$

$$\left(\frac{dy}{dx}\right)_v = \frac{-\left(\frac{\partial v}{\partial x}\right)}{\left(\frac{\partial v}{\partial y}\right)} = \frac{\left(\frac{\partial u}{\partial x}\right)}{\left(\frac{\partial u}{\partial x}\right)}$$

$$\left(\frac{dy}{dx}\right)_u = \frac{-1}{\left(\frac{dy}{dx}\right)_v} \Rightarrow dx_u dx_v + dy_u dy_v = 0$$

Derivative Rules

$$r = \sqrt{x^2+y^2}$$

$$\theta = \tan^{-1}\left(\frac{y}{x}\right)$$

$f(z) = z \Rightarrow f'(z) = 1$

$f(z) = z^n \Rightarrow f'(z) = n z^{n-1}$

Power series

Ex $f(z) = \ln z = \ln r + i(\theta + 2\pi n)$

$u = \ln r \qquad v = \theta + 2\pi n$

✓ $\frac{\partial u}{\partial x} = \frac{1}{r}\frac{\partial r}{\partial x} = \frac{x}{r^2} \qquad \frac{\partial v}{\partial y} = \frac{\partial \theta}{\partial y} = \frac{1/x}{1+(\frac{y}{x})^2} = \frac{x}{r^2}$

✓ $\frac{\partial v}{\partial x} = \frac{\partial \theta}{\partial x} = \frac{-y}{r^2} \qquad \frac{\partial u}{\partial y} = \frac{y}{r^2}$

Analytic where $r \neq 0$

$$\frac{df}{dz} = \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x} = \frac{x}{r^2} + i\frac{-y}{r^2}$$

$$= \frac{x - iy}{r^2} = \frac{z^*}{zz^*} = \frac{1}{z}$$

Point at ∞

$$w = \frac{1}{z}$$

ex $f(z) = z$ near $\infty = z$

$f(w) = \frac{1}{w}$ near $0 = w$

$$S_n = \sum_{j=0}^{n} f(\zeta_j)\,(z_{j+1} - z_j)$$

$$\lim |z_{j+1} - z_j| \to 0$$

$$\lim_{n\to\infty} S_n \longrightarrow \int_{z_0}^{z_0'} f(z)\,dz$$

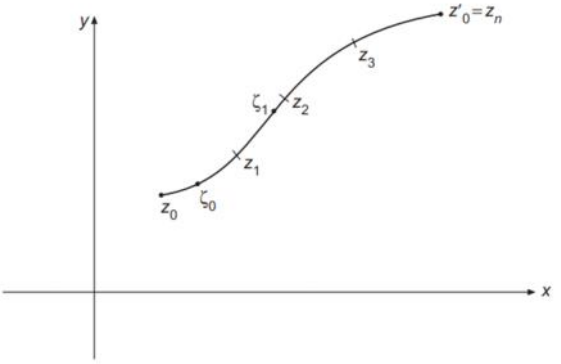


FIGURE 11.2 Integration path.

$$= \int_C f(z)\,dz$$

Contour Integral

$$= \int_C (u + iv)(dx + i\,dy) = \int_C (u\,dx - v\,dy) + i\int (v\,dx + u\,dy)$$

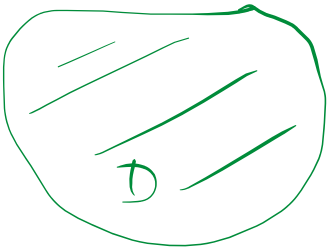
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Cauchy's Integral Thm

If $f(z)$ analytic, simply-connected region $\Rightarrow \oint_C f(z)\,dz = 0$

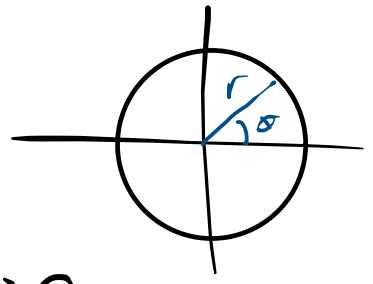
$$\int_{\partial D} \omega = \int_D d\omega$$

$a\,dx + b\,y$



Ex $f(z) = z^n$ $\quad C \equiv$ circle of r

$$\oint_C z^n \, dz$$



$z = re^{i\theta}$ $\qquad dz = re^{i\theta} \, i \, d\theta$

$$\oint r^n e^{in\theta} \, ir e^{i\theta} d\theta = ir^{n+1} \int_0^{2\pi} e^{i(n+1)\theta} d\theta$$

$$= ir^{n+1} \left. \frac{e^{i(n+1)\theta}}{i(n+1)} \right|_0^{2\pi} = 0 \quad \text{if } n \neq -1$$

$n = -1$

$$\oint \frac{1}{z}\, dz = \int_0^{2\pi} \frac{1}{r} e^{-i\alpha}\, i r e^{i\alpha}\, d\alpha$$

$$= i \int_0^{2\pi} d\alpha = 2\pi i$$

$$\oint_C z^n\, dz = \begin{cases} 0 & ; \ n \neq -1 \\ 2\pi i & ; \ n = -1 \end{cases}$$

Proof of Cauchy's Thm

$f = u + iv \qquad dz = dx + i\,dy$

$$\oint_C f(z)\,dz = \oint (u\,dx - v\,dy) + i \oint (v\,dx + u\,dy)$$

Stokes Thm: $$\oint_C (F\,dx + G\,dy) = \int_A \left(\frac{\partial G}{\partial x} - \frac{\partial F}{\partial y}\right) dx\,dy$$

$$\oint_C u\,dx - v\,dy = \int_A \left(-\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right) dx\,dy = 0$$

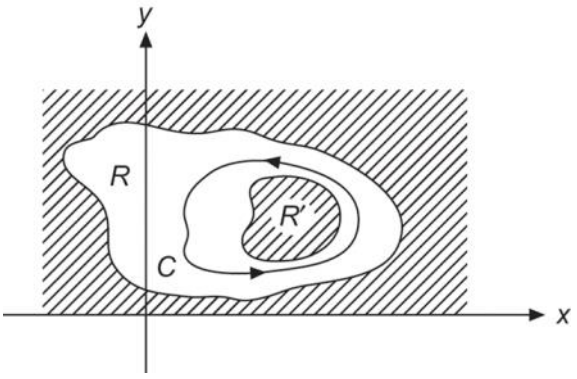
$$\oint_C (v\,dx + u\,dy) = \int_A \left(\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y}\right) dx\,dy = 0$$

Multiply connected region

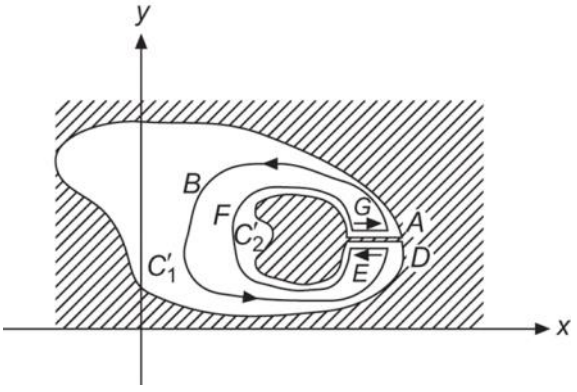
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$$\oint_C f(z)\,dz$$

$$\int_G^A = -\int_D^E$$



$$\oint_{C'} = \int_{ABD} + \int_{EFG} = \int_{C_1'} - \int_{C_2'} = 0$$



$$\oint_{C_1'} f(z)\,dz = \oint_{C_2'} f(z)\,dz$$

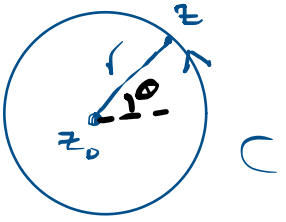
$$\oint_C (z-z_0)^n dz = 2\pi i \, \delta_{n,-1} = \begin{cases} 0, & n \neq -1 \\ 2\pi i, & n = -1 \end{cases}$$

Cauchy integral formula

$$\frac{1}{2\pi i} \oint_C \frac{f(z)}{z-z_0} dz = \begin{cases} f(z_0), & z_0 \text{ inside } C \\ 0, & z_0 \text{ outside } C \end{cases}$$

$z - z_0 = r e^{i\theta} \Rightarrow dz = r e^{i\theta} i d\theta$

$$\frac{1}{2\pi i} \int \frac{f(z_0 + r e^{i\theta})}{r e^{i\theta}} r e^{i\theta} i d\theta$$



$$\lim_{r \to 0} \frac{f(z_0)}{2\pi i} i \int d\theta = f(z_0$$

$$f(z_0) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{z - z_0} dz$$

C encircle z_0

$$f(z) = \frac{1}{2\pi i} \oint_C \frac{f(z')}{z' - z} dz'$$

Ex 11.4.1 $I = \oint_C \frac{dz}{z(z+2)}$ $C \equiv$ ccw unit circle center at $0 = z_0$

$$= \oint_C \frac{\frac{1}{z+2}}{z}\, dz = 2\pi i\, f(0) = \pi i$$

Ex 11.4.2 $I = \oint_C \frac{dz}{4z^2-1}$ $C \equiv$ ccw unit circle at 0

$$\frac{1}{4z^2-1} = \frac{1}{4}\left(\frac{1}{z-\frac{1}{2}} - \frac{1}{z+\frac{1}{2}}\right)$$

$$I = \frac{1}{4}\oint_C \frac{1}{z-\frac{1}{2}}\, dz - \frac{1}{4}\oint \frac{1}{z+\frac{1}{2}}\, dz = 0$$

$$f(z) = \frac{1}{2\pi i} \oint_C \frac{f(w)\,dw}{w - z}$$

Derivatives:

$$f^{(1)}(z) = \frac{1}{2\pi i} \oint_C \frac{f(w)\,dw}{(w - z)^2}$$

$$f^{(2)}(z) = \frac{2}{2\pi i} \oint_C \frac{f(w)}{(w-z)^3}\,dw$$

$$f^{(n)}(z) = \frac{n!}{2\pi i} \oint_C \frac{f(w)}{(w-z)^{n+1}}\,dw$$

$$\frac{1}{1-t} = \sum_{n=0}^{\infty} t^n \qquad t_0 = 0 \qquad |t| < 1$$

Ex $\quad \frac{1}{t}$ about $t_0 = 1$

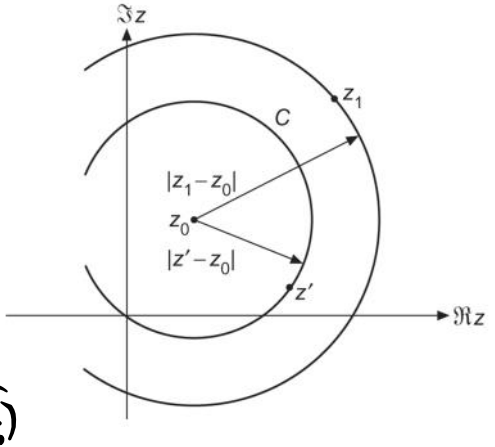
$$\frac{1}{t-1+1} = \frac{1}{1+(t-1)} = \sum_{n=0}^{\infty} (-1)^n (t-1)^n$$

$$|t-1| < 1$$

$$0 < t < 2$$

$f(z)$ about z_0 $\qquad |z-z_0| < |z'-z_0| < |z_1-z_0|$

$$f(z) = \frac{1}{2\pi i}\oint_C \frac{f(z')}{z'-z}\,dz'$$



$$= \frac{1}{2\pi i}\oint \frac{f(z')\,dz'}{(z'-z_0)-(z-z_0)}$$

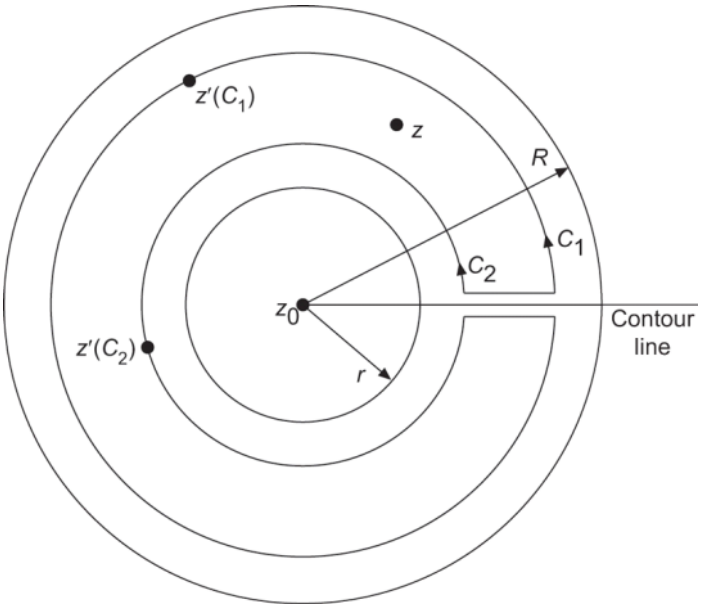
$$= \frac{1}{2\pi i}\oint \frac{1}{z'-z_0}\,\frac{f(z')\,dz'}{1-(z-z_0)/(z'-z_0)}$$

$$= \frac{1}{2\pi i}\oint \frac{1}{z'-z_0}\,f(z')\,dz' \sum_{n=0}^{\infty} \frac{(z-z_0)^n}{(z'-z_0)^n}$$

$$= \sum_{n=0}^{\infty} (z-z_0)^n \,\frac{1}{2\pi i}\oint \frac{f(z')\,dz}{(z'-z_0)^{n+1}} = \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!}(z-z_0)^n$$

Taylor series

$$f(z) = \frac{1}{2\pi i} \oint_C \frac{f(z')\,dz'}{z'-z}$$



$$= \frac{1}{2\pi i} \oint_{C_1} \frac{f(z')\,dz'}{z'-z} - \frac{1}{2\pi} \oint_{C_2} \frac{f(z')\,dz'}{z'-z}$$

$$= S_1 + S_2$$

$$S_1 = \frac{1}{2\pi i} \sum_{n=0}^{\infty} (z-z_0)^n \oint_{C_1} \frac{f(z')\,dz'}{(z'-z_0)^{n+1}}$$

z interior to C_1

$$S_2 = \frac{1}{2\pi i} \sum_{n=0}^{\infty} (z-z_0)^{-n} \oint_{C_2} \frac{f(z')\,dz'}{(z'-z_0)^{-n+1}}$$

z exterior to C_2

Laurent series:

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z - z_0)^n$$

$$a_n = \frac{1}{2\pi i} \oint \frac{f(z')\, dz'}{(z' - z_0)^{n+1}}$$

Ex $f(z) = 1 + \frac{z}{2} + \frac{z^2}{4} + \frac{z^3}{8} + \cdots$

$$+ \frac{2}{z} + 4\left(\frac{1}{z^2} - \frac{1}{z^3} + \frac{1}{z^4} - \cdots\right)$$

$z_0 = 0$

$|\frac{z}{2}| < 1 \quad (|z| < 2)$

&

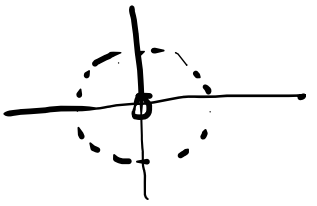
$|\frac{1}{z}| < 1 \quad (|z| > 1)$

$1 < |z| < 2$

Laurent Series

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Ex 11.5.1 $f(z) = \dfrac{1}{z(z-1)}$ $z_0 = 0$



$$f(z) = -\frac{1}{1-z} - \frac{1}{z} = -\frac{1}{z} - \sum_{n=0}^{\infty} z^n$$

$$= -\sum_{n=-1}^{\infty} z^n = \sum_{n=-\infty}^{\infty} a_n z^n$$

$$\frac{1}{2\pi i}\oint \frac{dz'}{(z')^{n+1}(z'-1)} = a_n = \begin{cases} -1 & ; n \geq -1 \\ 0 & ; n < -1 \end{cases}$$

$$\oint_C \frac{1}{(z')^{n+2}(z'-1)}\,dz' = \oint_C \frac{-1}{(z')^{n+2}} \sum_{m=0}^{\infty} (z')^m \,dz'$$

$$= -\sum_{m=0}^{\infty} \oint_C (z')^{m-n-2}\,dz'$$

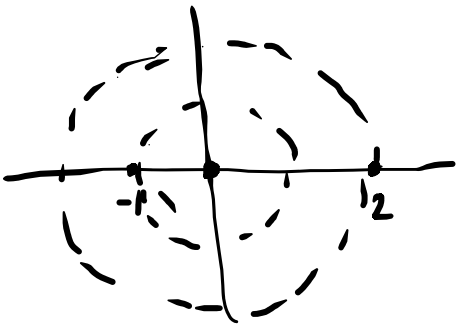
$(z')^{m-n-2} \to 0$ except $m-n-2 = -1$, $m = n+1$

$$= -\sum_{m=0}^{\infty} 2\pi i\, \delta_{m,n+1}$$

$$= \begin{cases} -2\pi i & n \geq -1 \\ 0 & n < -1 \end{cases}$$

Ex $f(z) = \dfrac{12}{z(2-z)(1+z)} = \dfrac{4}{z}\left(\dfrac{1}{1+z} + \dfrac{1}{2-z}\right)$

$0 < |z| < 1$



$$f(z) = \frac{4}{z}\left[\sum_{n=0}^{\infty}(-1)^n z^n + \frac{1}{2}\sum_{n=0}^{\infty}\left(\frac{z}{2}\right)^n\right]$$

$$= \frac{4}{z}\left[1 - z + \cdots + \frac{1}{2} + \frac{z}{2} + \cdots\right.$$

$|z|>2$

$$\frac{1}{1+z} = \frac{1}{z}\,\frac{1}{1+\frac{1}{z}} = \frac{1}{z}\sum_{n=0}^{\infty}\left(\frac{1}{z}\right)^n \qquad |z|>1$$

$$\frac{1}{2-z} = \frac{-1}{z}\,\frac{1}{1-\frac{2}{z}} = \frac{-1}{z}\sum_{n=0}^{\infty}\left(\frac{2}{z}\right)^n \qquad |z|>2$$

$1<|z|<2$

$$\frac{1}{1+z} = \frac{1}{z}\,\frac{1}{1+\frac{1}{z}} \qquad |z|>1$$

$$\frac{1}{2-z} = \frac{1}{2}\,\frac{1}{1-z/2} \qquad |z|<2$$

$\Rightarrow\ 1<|z|<2$

11.6 Singularity

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Isolated singular Points

z_0

Laurent expansion about z_0

1) most negative power $(z-z_0)^{-n}$ $\Rightarrow$ "Pole of order n"

2) negatively infinite power "essential sing."

simple pole $\equiv$ pole of order 1

Isolated sing

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To identify pole

$\lim_{z \to z_0} (z - z_0)^n f(z) \Rightarrow$ smallest n for which limit exists.

Ex $f(z) = \frac{-3}{z^2} + \frac{1}{z} + 1 + 2z + \cdots$

Pole of order 2

Ex $e^{1/z} = \sum_{n=0}^{\infty} \frac{1}{n!} \left(\frac{1}{z}\right)^n$

essential singularity

Branch points

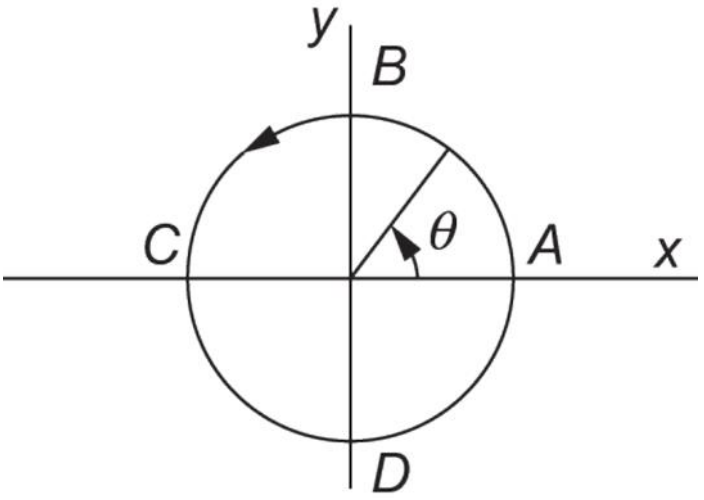
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Associated with multivalued functions

Ex $z^{1/2} = f(z)$

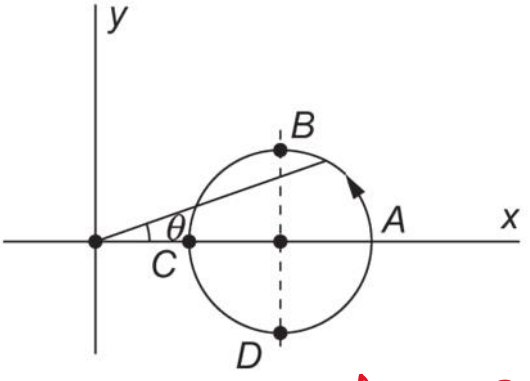
stat $f(1) = +1$

$z^{1/2} = e^{i\theta/2}$



Ex $z^{1/2}$

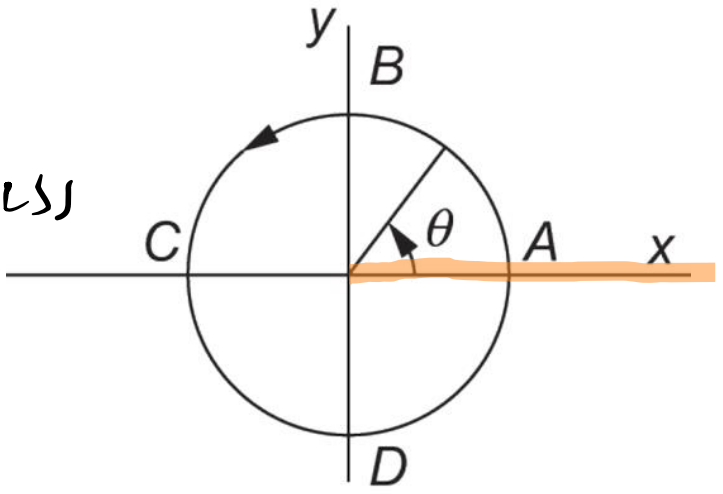
A → $f(z) = \sqrt{3}$



$z_0 = 0$ is a branch point of order 2

Branch line (Cut)

To prevent multivaluedness



Ex $f(z) = z^{1/2}$

has two branches

$f_+(z) = r^{1/2} e^{i\theta/2}$

$f_-(z) = -r^{1/2} e^{i\theta/2}$

Ex $\ln z = \ln r + i(\alpha + 2\pi n)$

Branch point $z_0 = 0$

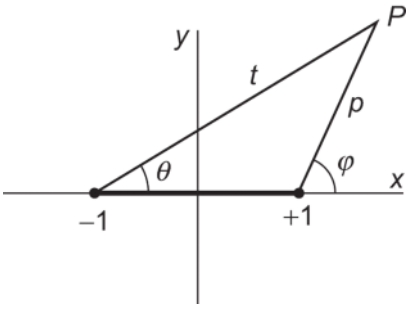
of order ∞

Ex $f(z) = (z^2-1)^{1/2} = (z-1)^{1/2}(z+1)^{1/2}$

Branch points $z = -1, +1$

$z-1 = \rho e^{i\phi}$ $\qquad z+1 = r e^{i\theta}$

$f(z) = (r\rho)^{1/2} e^{i(\theta+\phi)/2}$



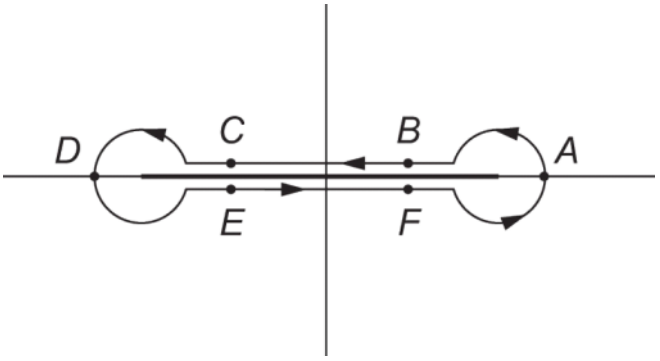
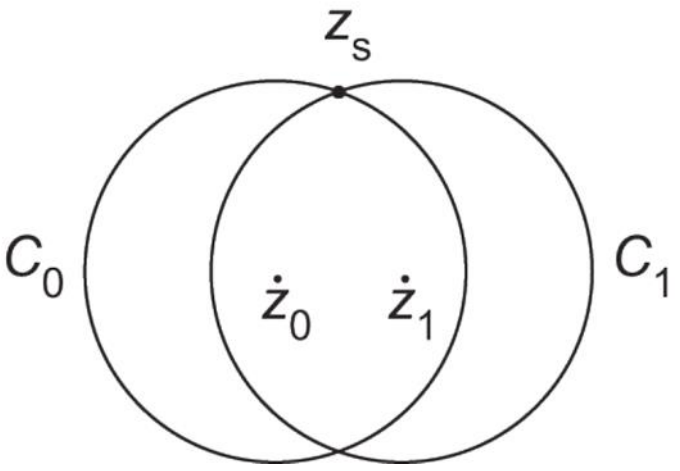


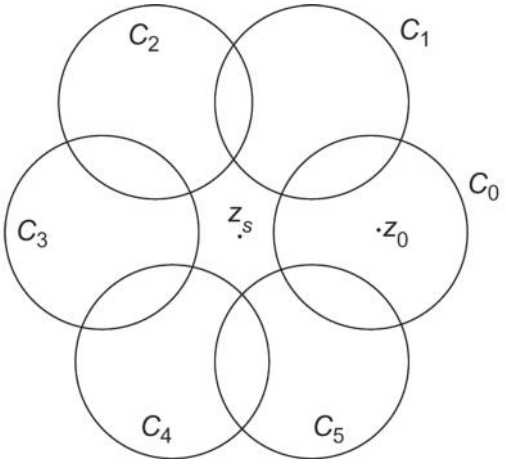
Table 11.1 Phase Angles, Path in Fig. 11.13

Point	θ	φ	$(\theta+\varphi)/2$
A	0	0	0
B	0	π	$\pi/2$
C	0	π	$\pi/2$
D	π	π	π
E	2π	π	$\pi/2$
F	2π	π	$3\pi/2$
A	2π	2π	2π

Analytic Continuation

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$f(z)$ analytic in A

$\tilde{f}(z)$ analytic in B

$A \subset B$

$\tilde{f}$ analytic contin.

Ex $f(z) = 1 + z + z^2 + \cdots$

$|z| < 1$

$\tilde{f}(z) = \dfrac{1}{1-z}$

The Residue Thm

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$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z - z_0)^n$$

$\oint_C f(z)\,dz$ $\quad C$ encircles isolated singularity z_0

$$a_n \oint (z - z_0)^n \, dz = 0 \; ; \; n \neq -1$$

$$a_n \oint (z - z_0)^n \, dz = 2\pi i \, a_{-1} \; ; \; n = -1$$

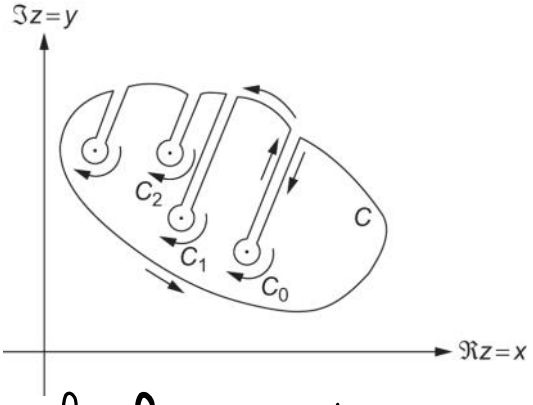
$a_{-1} \equiv$ residue of $f(z)$ at $z = z_0$

Residue Thm

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$$\oint f(z)\,dz = \int_C f(z)\,dz + \sum_i \int_{C_i} f(z)\,dz = 0$$

$$\oint_{C_i} f(z)\,dz = -2\pi i\, a_{-1,i}$$



$$\oint_C f(z)\,dz = 2\pi i \sum_i a_{-1,i}$$

$$= 2\pi i \sum_i \text{residues of } f(z) \text{ at } z_i$$

Computing Residue

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Simple pole

$$(z-z_0)f(z) = a_{-1} + a_0(z-z_0) + a_1(z-z_0)^2 + \cdots$$

$$a_{-1} = \lim_{z \to z_0} (z-z_0) f(z)$$

ex $f(z) = \dfrac{1}{z^2+1} = \dfrac{1}{(z-i)(z+i)}$

$z = -i$ $\lim_{z \to -i} (z+i) \dfrac{1}{z^2+1} = \lim_{z \to -i} \dfrac{1}{z-i} = \dfrac{-1}{2i} = \dfrac{1}{2}i$

$z = i$ $\lim_{z \to i} (z-i) \dfrac{1}{z^2+1} = -\dfrac{1}{2}i$

Computing Resi

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nth Pole

$$(z-z_0)^n f(z) = a_{-n} + a_{-n+1}(z-z_0)^1 + \cdots + a_{-1}(z-z_0)^{n-1} + a_0(z-z_0)^n + \cdots$$

$$a_{-1} = \frac{1}{(n-1)!}\lim_{z\to z_0}\left\{\frac{d^{n-1}}{dz^{n-1}}\left[(z-z_0)^n f(z)\right]\right\}$$

Examples

Res. of $\frac{1}{4z+1}$ at $z = \frac{-1}{4} \Rightarrow \lim_{z\to -\frac{1}{4}} \frac{z+\frac{1}{4}}{4z+1} = \frac{1}{4}$

* Resid. of $\frac{1}{\sin z}$ at $z=0 \Rightarrow \lim_{z\to 0} \frac{z}{\sin z} = 1$

* Res. of $\frac{z}{\sin^2 z}$ at $z=\pi$ (2nd order pole)

$$\lim_{z\to\pi} \frac{d}{dz}\left[\frac{(z-\pi)^2 z}{\sin^2 z}\right] \qquad \omega = z-\pi$$

$$= \lim_{\omega\to 0} \frac{d}{dz}\left[\frac{\omega^2(\omega+\pi)}{\sin^2\omega}\right] \quad ; f(\omega) = \frac{(\omega+\pi)}{\left[\omega^2 - \frac{\omega^4}{3} + \cdots\right]}$$

$$f(\omega) = \frac{(\omega+\pi)}{\left[\omega^2 - \frac{\omega^4}{3} + \cdots\right]} = \frac{\omega+\pi}{\omega^2\left[1 - \frac{\omega^2}{3} + \cdots\right]}$$

$$\simeq \frac{\omega+\pi}{\omega^2} + \frac{\omega+\pi}{1 - \frac{\omega^2}{3} + \cdots}$$

Residue is $\underline{1}$

Ex $f(z) = e^{-1/z}$ essential singularity at $z = 0$

$$e^{-1/z} = 1 - \frac{1}{z} + \frac{1}{2}\frac{1}{z^2} - \frac{1}{6}\frac{1}{z^3} + \cdots$$

Residue of $f(z)$ at $z = 0$ is -1

Singularity on Contour

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Divergence

Ex $\int_{-a}^{b} \frac{1}{x}\,dx$ singularity at $x=0$

Can be modified $\Rightarrow \lim_{\delta\to 0}\left\{\int_{-a}^{-\delta}\frac{dx}{x}+\int_{\delta}^{b}\frac{dx}{x}\right\}$!!!

$\ln b - \ln a$

$\lim_{\delta_1,\delta_2\to 0^+}\left[\int_{-a}^{-\delta_1}\frac{dx}{x}+\int_{+\delta_2}^{b}\frac{dx}{x}\right]$ $\quad \delta_2 = 2\delta_1$

$\ln b - \ln a - \ln 2$

Cauchy's Principal Value

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$$\lim_{\delta \to 0^+} \int_{x+\delta}^{x_0-\delta} f(x)\,dx + \int_{x+\delta}^{} f(x)\,dx \qquad "x_0 \equiv \text{sing.}"$$

$$P\int f(x)\,dx \qquad \oint f(x)\,dx$$

Ex 11.7.2

$$\int_0^\infty \frac{\sin x}{x}\,dx = \int_0^\infty \frac{e^{ix} - e^{-ix}}{2ix}\,dx$$

$$I = \int_{-\infty}^{\infty} \frac{e^{ix}}{2ix}\,dx$$

$$= \int_{-\infty}^{0} \frac{e^{ix}}{2ix}\,dx + \int_{}^{\infty} \frac{e^{ix}}{2ix}\,dx$$

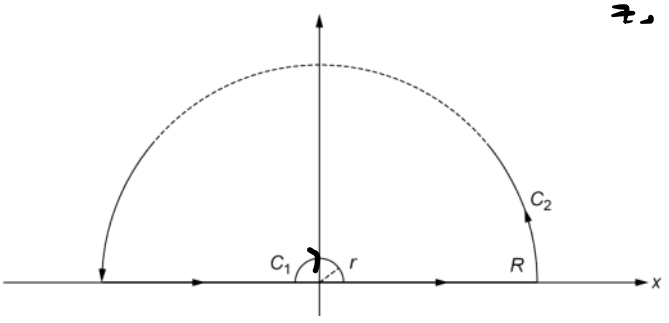
Cauchy P. Value

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$$\oint_C f(z)dz = \int f(x)dx + I_{over} + \int_{C_2} f(z)dz = 2\pi i \sum \text{Resid}$$

"other than z_0"

$$I_{over} = -\pi i\, a_{-1}$$



11.8 Definite integrals

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Trig. Integrals, $(0, 2\pi)$

$$I = \int_0^{2\pi} f(\sin\theta, \cos\theta)\, d\theta$$

$z = e^{i\theta}$

$dz = i e^{i\theta} d\theta$

$= iz\, d\theta \Rightarrow$ $\boxed{d\theta = -i \frac{dz}{z}}$

$$\sin\theta = \frac{z - z^{-1}}{2i}, \quad \cos\theta = \frac{z + z^{-1}}{2}$$

$\Rightarrow \oint_C f(z)\, dz$ $\quad C \equiv$ unit circle

Ex $\quad I = \int_0^{2\pi} \frac{d\theta}{1 + a\cos\theta} \qquad |a| < 1$

$$I = \oint_C \frac{-i\,dz}{z\left[1 + (a/2)(z + z^{-1})\right]} = -i \oint_C \frac{dz}{z + \frac{a}{2}(z^2 + 1)}$$

$$= -\frac{2i}{a} \oint_C \frac{dz}{z^2 + \frac{2}{a}z + 1} \qquad z_1 z_2 = 1$$

$$-\frac{2}{a} \mp \sqrt{\left(\frac{2}{a}\right)^2 - 4} \Rightarrow \quad z_1 = -\frac{1 + \sqrt{1 - a^2}}{a}, \; z_2 = -\frac{1 - \sqrt{1 - a^2}}{a}$$

Residue at $z_2 \Rightarrow$ $\lim_{z \to z_2} \dfrac{(z - z_2)}{(z - z_1)(z - z_2)}$

$= \dfrac{1}{z_2 - z_1} = \dfrac{-a}{2\sqrt{1-a^2}}$

$I = -\dfrac{2i}{a}(2\pi i) \dfrac{a}{2\sqrt{1-a^2}} = \dfrac{2\pi}{\sqrt{1-a^2}}$

Integrals, range $-\infty$ to $+\infty$

$$I = \int_{-\infty}^{\infty} f(x)\,dx$$

$f(z)$ vanishes faster than $\frac{1}{z}$

Thm if $\lim_{z\to\infty} z f(z) = 0$ for all $z = Re^{i\alpha}$ $(\alpha_1 \le \alpha \le \alpha_2)$

$$\lim_{R\to\infty} \int_C f(z)\,dz = 0$$



C : arc with $\alpha_1 \le \alpha \le \alpha_2$, center at origin, radius R

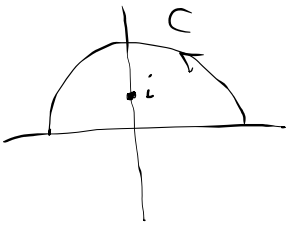
Proof

$$\lim_{R\to\infty}\left|\int_C f(z)\,dz\right| \le \lim_{R\to\infty}\int |f(z)\,dz|$$

$$\le \lim_{R\to\infty}\int_{\alpha_1}^{\alpha_2} |f(Re^{i\alpha})\, iRe^{i\alpha}\,d\alpha|$$

$$\le (\alpha_2-\alpha_1)\lim_{z\to\infty} f(z)z = 0$$

Ex $\quad I = \int_0^\infty \frac{dx}{1+x^2} = \frac{1}{2}\int_{-\infty}^{\infty} \frac{dx}{1+x^2}$



$$\oint \frac{dz}{1+z^2} = 2\pi i \left(\text{Residue of } f(z) \text{ at } i\right)$$

$$= 2\pi i \lim_{z \to i} \frac{z-i}{1+z^2} = \pi$$

$$= \int_C + \int_{-\infty}^{\infty} \frac{dx}{1+x^2}$$

$$I = \frac{1}{2}\pi$$

Integrals with complex exponentials

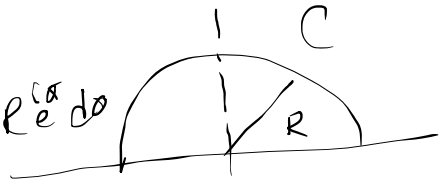
$$I = \int_{-\infty}^{\infty} f(x)\, e^{iax}\, dx \qquad \text{real } a>0$$

* $f(z)$ analytic

* $\lim_{|z|\to\infty} f(z) = 0$

$0 \le \arg z \le \pi$

$$\int_C f(z)\, e^{iaz}\, dz = \int_0^{\pi} f(Re^{i\theta})\, e^{ia(R\cos\theta + iR\sin\theta)}\, iRe^{i\theta}\, d\theta$$



$\lim_{R\to\infty} \to 0$

Jordan's Lemma

If $\lim_{z\to\infty} f(z) = 0$ for all $z = Re^{i\theta}$ $0 \le \theta \le \pi$

$$\lim_{R\to\infty} \int_C f(z) e^{iaz}\, dz = 0 \qquad a>0$$

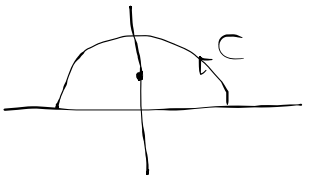
Ex 11.8.4

$$I = \int_0^\infty \frac{\cos x\, dx}{x^2+1} = \frac{1}{2}\left[\int_0^\infty \frac{e^{ix}}{x^2+1}dx + \int_0^\infty \frac{e^{-ix}}{x^2+1}dx\right]$$

$$= \frac{1}{2}\int_{-\infty}^{\infty} \frac{e^{ix}}{x^2+1}\, dx$$

$$\oint \frac{e^{iz}}{z^2+1}\,dz = 2\pi i \cdot \text{Resid at } z=i$$

$$= 2\pi i \lim_{z\to i} \frac{e^{iz}}{z+i} = \frac{\pi}{e}$$



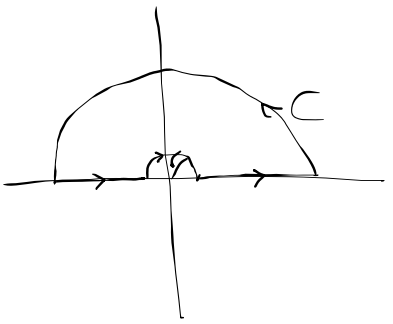
$$= \int_C + \int_{-\infty}^{\infty} \frac{e^{ix}}{x^2+1}\,dx \quad \Rightarrow \quad \int_{-\infty}^{\infty} \frac{e^{ix}}{x^2+1}\,dx = \frac{\pi}{e}$$

Jordan's Lemma

$$\int_0^{\infty} \frac{\cos x}{x^2+1}\,dx = \frac{\pi}{2e}$$

Ex 11.8.5

$$I = \int_0^\infty \frac{\sin x}{x}\,dx = \int_{-\infty}^{\infty} \frac{e^{ix}}{2ix}\,dx$$



$$\oint \frac{e^{iz}}{2iz}\,dz = 0 = \int_r + \int_C + \int_{-\infty}^{\infty} \frac{e^{ix}}{2ix}\,dx$$

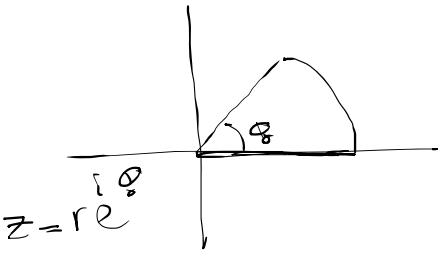
$$0 = -\pi i\,(\text{Resid at } z=0) + I$$

$$0 = -\pi i\left(\frac{1}{2i}\right) + I \Rightarrow \quad I = \frac{\pi}{2}$$

Other technique

range $(0, \infty)$ no symmetry

Ex 11.8.6 $\quad I = \int_0^\infty \frac{dx}{x^3+1}$



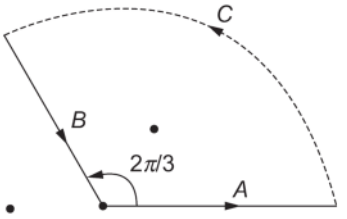
$z = re^{i\theta}$

$z^3 = r^3 \qquad \theta_0 = \frac{2\pi}{3}$

$$\oint \frac{dz}{z^3+1} = 2\pi i \,(\text{Resid at } z = z_1)$$

$z_1 \quad z_2 \quad z_3$

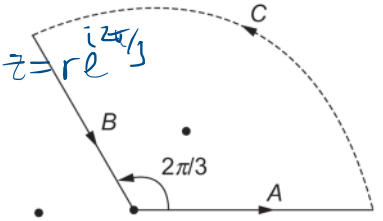
$(-1)^{1/3} = e^{i(\pi+2n\pi)/3} = e^{i\pi/3},\ e^{i\pi},\ e^{i\frac{5\pi}{3}}$



$z^3+1 = 0 \Rightarrow z^3 = -1$

$$\text{Resid at } z = z_1 = \lim_{z \to z_1} \frac{z - z_1}{z^3+1} = \lim_{z \to z_1} \frac{1}{3z^2} = \frac{1}{3e^{i2\pi/3}}$$

$$\oint \frac{dz}{z^3+1} = \frac{2\pi i}{3e^{i2\pi/3}} = I + \int_C + \int_B$$



$$\int_B \frac{dz}{z^3+1} = e^{i2\pi/3} \int_\infty^0 \frac{dr}{r^3+1} = -e^{i2\pi/3} I$$

$$\left[\frac{2\pi i}{3e^{i2\pi/3}} = I\left(1 - e^{i2\pi/3}\right) \right] e^{-i\pi/3}$$

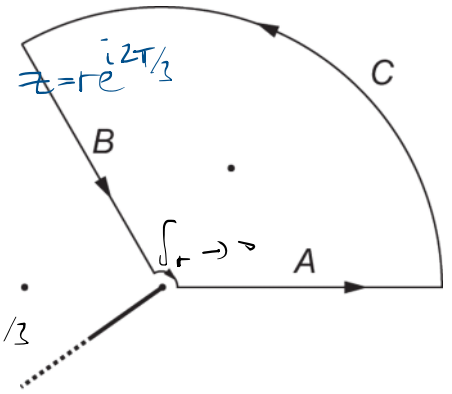
$$\frac{2\pi i}{3e^{i\pi}} = I\left(e^{-i\pi/3} - e^{i\pi/3}\right) \Rightarrow \frac{\pi}{3} = I \sin\frac{\pi}{3}$$

$$I = \frac{\pi/3}{\sin(\pi/3)}$$

Avoidance of Branch Points

Principal Branch
$\ln z = \ln r + i\theta$

Ex 11.87 $I = \int_0^\infty \frac{\ln x\, dx}{x^3+1}$



$$\oint \frac{\ln z\, dz}{z^3+1} = 2\pi i \text{ Resid at } z_1 =$$

$$2\pi i \lim_{z\to z_1} \frac{(z-z_1)\ln z}{z^3+1} = \frac{\pi i}{9} e^{-2\pi i/3}$$

$$e^{i2\pi/3}\int_\infty^0 \frac{(\ln r + i\frac{2\pi}{3})}{r^3+1} dr = -e^{i2\pi/3}\left(I + i\frac{2\pi}{3}\int_0^\infty \frac{dr}{r^3+1}\right)$$

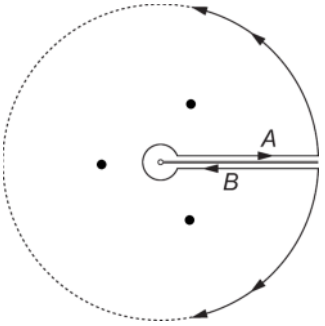
$$\int_0^{2\pi/3} \frac{\ln r}{r^3+1}(da\, i^0$$

$$2\pi i \frac{\pi i}{9} e^{-2\pi i/3} = I - e^{2\pi i/3} I - \frac{2\pi i}{3}\frac{\pi/3}{\sin \pi/3} \Rightarrow I = -\frac{2\pi^2}{27}$$

Ex 11.8.9 Exploiting Branch cut

Wednesday, January 28, 2026 4:15 PM

$$I = \int_0^\infty \frac{dx}{x^3+1}$$ use $$\oint \frac{\ln z\, dz}{z^3+1}$$ on the contour



$$\oint \frac{\ln z\, dz}{z^3+1} = 2\pi i \sum \text{Resid} = 2\pi i \left(R_1+R_2+R_3\right)$$

$$= \int_A + \int_B$$

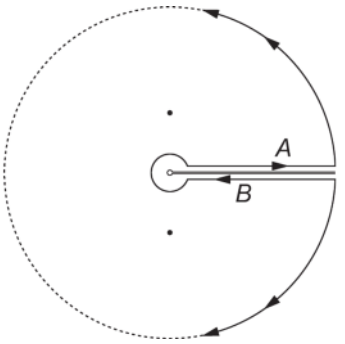
$$\int_0^\infty \frac{\ln x\, dx}{x^3+1} + \int_\infty^0 \frac{\ln x + 2\pi i}{x^3+1}\, dx = -2\pi i\, I$$

$$\Rightarrow \quad I = -\left(R_1+R_2+R_3\right) = \frac{\pi/3}{\sin \pi/3}$$

Ex 11.8.8 Exploiting Branch Cut

Wednesday, January 26, 2026 4:59 PM

$$I = \int_0^\infty \frac{x^p\,dx}{x^2+1}, \qquad 0 < p < 1$$



$$\oint \frac{z^p\,dz}{z^2+1} = 2\pi i\,(R_1+R_2) = 2\pi i\left(\frac{e^{ip\pi/2}}{2i} - \frac{e^{i3p\pi/2}}{2i}\right)$$

$$= \int_A \quad + \int_B$$

$$= I - e^{2p\pi i} I \qquad \Rightarrow \quad I\left(1 - e^{2p\pi i}\right) = 2\pi i \left(\frac{e^{ip\pi/2} - e^{i3\pi p/2}}{2i}\right)$$

$$\Rightarrow \qquad I = \pi\,\frac{\sin p\pi/2}{\sin p\pi}$$